Ch 1. Divisibility

Def. $a, b \in \mathbb{Z}$. $a \neq 0$. b is divisible by a if
$\exists x \in \mathbb{Z}$ s.t. $b = ax$. Write $a \mid b$.
(More notation: $a \nmid b$, $a^3 \| b \Leftrightarrow a^3 \mid b$ and $a^4 \nmid b$)

Properties
① A partial order
i) reflexive : $a \mid a$
ii) anti-symmetric : $a \mid b$ and $b \mid a \Rightarrow a = \pm b$
iii) transive : $a \mid b$ and $b \mid c \Rightarrow a \mid c$
(cf: equivalence class : reflexive, symm, transitive e.g. "=")
② Arithmetic :
i) $a \mid b$, $b \mid c \Rightarrow a \mid (bx + cy)$ $\forall x, y \in \mathbb{Z}$
ii) $a \mid b \Rightarrow ma \mid mb$ $(m \neq 0)$
③ Size: $a \mid b$ and $a, b > 0 \Rightarrow a \leq b$

Thm 1.2 (Division Alg) Given $a, b \in \mathbb{Z}$, $a > 0$, $\exists!$ q and $r \in \mathbb{Z}$ s.t.
$b = qa + r$, $0 \leq r < a$

pf) Idea: b에서 a를 $\pm$하다보면 $[0, a)$ 구간에 도달.
$S := \{ x \in \mathbb{Z}^{\geq 0} \mid x = b - aq,\ q \in \mathbb{Z} \}$
Since S is nonempty, it has a minimum $r \geq 0$.
Moreover, $r < a$. Otherwise $r - a \in S$ □

- "Algorithm" is a mathemafical procedure or method to obtain a result. e.g. Division Alg.

Def. a is a common divisor of b and c if $a|b$ and $a|c$.

The $\gcd(a,b)$ or (a,b) is the greatest one among common divisors

Thm 1.3. Let $g=(b,c)$. $\exists\, x_0, y_0 \in \mathbb{Z}$ s.t $g = bx_0+cy_0$.

More precisely, $g = \min\{bx+cy>0 \mid x,y\in\mathbb{Z}\}$

pf) $\ell := \min\{bx+cy>0 \mid x,y\in\mathbb{Z}\}$. say $\ell = bx_0+cy_0$. ETS $\ell=(b,c)$

$\because S$

Claim: $\ell \mid b$ $\quad 0\le r<\ell$

(∵) Supp $b = \ell q + r$, $r>0$. $r = b-\ell q = b(1-x_0 q) + c(-y_0 q) \in S$

$0<r<\ell$ $\rightarrow\leftarrow$ $\quad\therefore r=0$ and $\ell \mid b$ ///

∵ $\ell \mid b,c \Rightarrow \ell$ is a common divisor. $\therefore \ell \le g$.

Since $g \mid b,c$, $g \mid bx_0+cy_0 = \ell$ $\quad\therefore g \le \ell$

$\Rightarrow g=\ell$ □

Question: ① $\exists\, x_1,\cdots,x_n \in \mathbb{Z}$ s.t. $(b_1,b_2,\cdots,b_n) = \sum_{i=1}^{n} b_i x_i$

② How to compute (a,b)?

③ " " $x,y\in\mathbb{Z}$ s.t. $ax+by=(a,b)$?

Cor. $(ma,mb) = m(a,b)$, $m>0$.

HW $(a,m)=(b,m)=1 \Rightarrow (ab,m)=1$.

Cor. (Euclid Lemma) If $c|ab$ and $(b,c)=1$, then $c|a$.

pf) Let $bx+cy=1$. $c \mid \underline{abx} + \underline{acy} = a(bx+cy) = a$ □

Thm 1.11 (Euclid Alg). Given $a, b \in \mathbb{Z}$, $b > 0$, we have

$$a = bq_1 + r_1, \quad 0 < r_1 < b$$
$$b = r_1 q_2 + r_2, \quad 0 < r_2 < r_1$$
$$\vdots$$
$$r_{j-2} = r_{j-1} q_j + r_j, \quad 0 < r_j < r_{j-1} \quad \Rightarrow r_j = r_{j-2} - r_{j-1} q_j$$
$$r_{j-1} = r_j q_{j+1}$$

① $(a, b) = (b, r_1) = \cdots = (r_{j-1}, r_j) = r_j$

② $\begin{pmatrix} r_{j-1} \\ r_j \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -q_j \end{pmatrix} \begin{pmatrix} r_{j-2} \\ r_{j-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -q_j \end{pmatrix} \cdots \begin{pmatrix} 0 & 1 \\ 1 & -q_1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}$ ($r_1 = a$, $r_0 = b$)

$M_j := \begin{pmatrix} s_{j-1} & t_{j-1} \\ s_j & t_j \end{pmatrix}$, $M_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$$\begin{pmatrix} r_{j-1} \\ r_j \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -q_j \end{pmatrix} \begin{pmatrix} * & * \\ s_{j-1} & t_{j-1} \end{pmatrix} = \begin{pmatrix} s_{j-1} & t_{j-1} \\ * & * \end{pmatrix}$$

Then $r_j = s_j a + t_j b$.

Q. ① Does it terminate? How many steps?

Yes (두번⇒한비트↓) $\quad \lambda \le \frac{\log b}{\log \phi} + 1$, $\phi = 1.62\cdots$

② Computational cost? $O(\log a \cdot \log b)$ bit operations.

Computational Cost.

Supp we have logic gates $f_\wedge(a,b) = ab \bmod 2$, $f_\oplus(a,b) = a+b \bmod 2$

$f_\neg(a) = 1 - a \bmod 2$.

Full Adder $(a, b, c) = \left(a+b+c \bmod 2, \left\lfloor \frac{a+b+c}{2} \right\rfloor \right)$ e.g. $a=b=c=1$, $a+b+c = (11)_2$

Let $a = (a_{n-1} \cdots a_0)_2$, $b = (b_{n-1} \cdots b_0)_2$ $a_i, b_i \in \{0, 1\}$

$$\begin{array}{r} c_n \; c_{n-1} \; c_2 \; c_1 \\ a_{n-1} \cdots a_1 \; a_0 \\ + \; b_{n-1} \cdots b_1 \; b_0 \\ \hline d_n \; d_{n-1} \cdots d_3 \; d_2 \; d_1 \end{array}$$

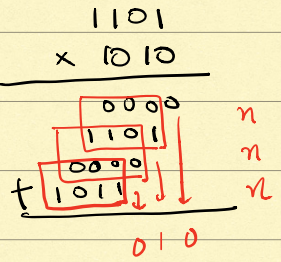
n번의 FA. ⇒ n

$$\begin{array}{r} 1000 \\ 1101 \\ + \; 1010 \\ \hline 10111 \end{array}$$

- Linear Complexity : Add, Subtract $O(n)$
- Quadratic Complexity : Mul, Division $(a \bmod b)$, EA

Mul $O(\log a \log b)$

EEA $O(\log b \cdot \log 8)$



$(n-1)$번 n 비트 덧셈 $= n(n-1)$ FA $= O(n^2)$

EA : 나눗셈을 $(j+1)$번 수행 $= O(\log^2 b \log 8)$?

$\log b \log 8 \leq 2 \log b$

$\Rightarrow O(\log a \log b)$

- Cubic Complexity : $a^b \bmod c$. a, b, c : n-bits

AB for $A, B \in M_n(\mathbb{Z})$ max entry : n-bits

- Exponential Complexity : 인수분해

w.r.t. input bit size

HW. §1.2. 2, 13, 16, 24, 43, 49, 51.